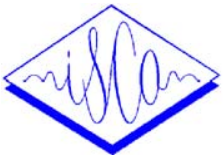
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ON BINARY AND RATIO TIME-FREQUENCY MASKS FOR ROBUST SPEECH RECOGNITION

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Abstract

A time-varying Weiner filter extracts the speech signal from a noisy mixture using the *a priori* signal-to-noise ratio in a local time-frequency unit. We estimate this ratio using a binaural processor and derive a ratio time-frequency mask. This mask is used to extract the speech signal, which is then fed to a conventional speech recognizer operating in the cepstral domain. We compare the performance of this system with a missing data recognizer that operates in the spectral domain using the time-frequency units dominated by speech. For use by the missing data recognizer, the same processor is used to estimate an ideal time-frequency binary mask, which selects the speech signal if it is stronger than the interference in a local time-frequency unit. We find that the performance of the missing data recognizer is better on a small vocabulary recognition task but the performance of the conventional recognizer is substantially better when the vocabulary size is larger.

1. Introduction

The performance of automatic speech recognizers (ASRs) degrades quickly in the presence of noise and other distortions [1]. To mitigate the effect of noise on recognition, noisy speech is typically preprocessed by speech enhancement algorithms, such as computational auditory scene analysis (CASA) systems (e.g. [2]) or spectral subtraction based systems (e.g. [3]). If samples of the corrupting noise source are available *a priori*, a model for the noise can additionally be trained and noisy speech may be jointly decoded based on the models of speech and noise [4]. However, in many realistic applications, the performance of the above approaches to robust speech recognition is inadequate [5].

Recently a missing data approach to robust speech recognition has been proposed [5]. This method is based on distinguishing between reliable and unreliable data. When speech is contaminated by additive noise, some time-frequency (T-F) regions will contain predominantly speech energy (reliable) and the rest are dominated by noise energy. The missing data method treats the latter T-F units as missing or unreliable during recognition (see Section 3.2). Missing T-F units are typically identified using spectral subtraction. The performance of missing data recognizer is significantly better than the performance of a system using spectral subtraction for speech enhancement followed by recognition of enhanced speech on a small vocabulary task [5].

The principal disadvantage of the missing data recognizer is that recognition is performed in the spectral or T-F domain. It is well known that recognition using cepstral coefficients yields a superior performance compared to recognition using spectral coefficients under clean speech conditions [6]. Attempts to adapt the missing data method to the cepstral domain have centered around reconstruction or imputation of the missing values in the spectral domain followed by transformation to the cepstral domain [5, 7]. This reconstruction is typically based either on the speech recognizer itself or on other trained models of speech. The success of these model-based imputation techniques depend on the adequacy of reliable data for identification of the correct speech model for imputation. In addition, errors in imputation procedures affect the performance of the system even when the model is correctly identified.

Another potential drawback of the missing data recognizer, which has not been well studied, is the problem of data paucity. At low SNRs, paucity of reliable data constrains the missing data recognizer to use only a small portion of the total T-F acoustic model space. This reduced space may be insufficient to differentiate between a large number of competing hypotheses during decoding. In this paper, we study this issue by comparing the performance of the missing data recognizer on two tasks with different vocabulary sizes.

We employ a recent binaural speech segregation system [8] as a front-end for the missing data recognizer. This system was chosen for its ability to estimate an ideal binary mask accurately. A T-F unit in the ideal binary mask is labeled reliable if the corresponding T-F unit of the noisy speech contains more speech energy than interference energy; it is labeled unreliable otherwise. This mask is fed to the missing data recognizer and recognition is performed in the spectral domain.

A frame-based Weiner filter, which utilizes *a priori* SNR in a local T-F unit, has been used previously to effectively enhance noisy speech [9]. It has been shown that in a narrow frequency band, there exists a systematic relationship between *a priori* SNR and values of the binaural cues of interaural time differences (ITD) and interaural intensity differences (IID) [8]. Motivated by this observation, we estimate an ideal ratio T-F mask using statistics collected for ITD and IID at each frequency. A unit in the ratio mask is a measure of the speech energy to total energy (speech and noise) in the corresponding T-F unit of noisy signal. This ratio mask is then used to resynthesize the speech signal, enabling recognition using Mel-Frequency Cepstral Coefficients (MFCCs). We use "conventional recognizer" to refer to a continuous density hidden Markov model (HMM) based ASR using MFCCs as features.

We compare the performance of the conventional recognizer with a missing data recognizer on a robust speech recognition task. In particular, we examine the effect of vocabulary size on the performance of the two recognizers. We find that on a small vocabulary task, the missing data recognizer outperforms the conventional ASR. Our finding is consistent with a previous comparison using a binaural front-end on a small vocabulary "cocktail-party" recognition task [10, 11]. With increase in vocabulary size, however, the conventional ASR performs substantially better. Results using the missing value imputation methods have been reported on a larger vocabulary previously [7]. Their method uses a binary mask and therefore is subject to the same limitations stated previously.

Section 2 describes the architecture and details of the binaural front-end. The conventional and missing data recognition methods are reviewed in Section 3. The two recognizers are tested on two different task domains with different vocabulary sizes. Section 4 presents the evaluation results of the recognizers along with a comparison of their relative performance. Finally, conclusions and future work are given in Section 5.

2. A localization based front-end for ASR

We employ a binaural front-end as a preprocessor for both the conventional and missing data recognizers. The input to the binaural system is a mixture of speech and interference presented at different, but fixed, locations. Signals are upsampled from their original frequencies (see Section 4) to 44.1 kHz. Binaural signals are obtained by filtering the monaural signals with measured head-related transfer functions (HRTFs) from a KEMAR dummy head. HRTFs provide location-dependent ITD and IID which can be extracted independently in each T-F unit. The T-F resolution is 20 ms time frames with a 10 ms frame shift, and 512 DFT coefficients. Frames are extracted by applying a running Hamming window to the signal. The ITD/IID estimates are based on the spectral ratio at the two ears:

$$\left(I\hat{T}D, I\hat{I}D\right)(\omega,t) = \left[-\frac{1}{\omega}\angle\frac{X_L(\omega,t)}{X_R(\omega,t)}, \frac{|X_L(\omega,t)|}{|X_R(\omega,t)|}\right] \quad (1)$$

where $X_L(\omega,t)$ and $X_R(\omega,t)$ are the left and right ear spectral values of the noisy speech signal at frequency ω and time t. An ideal ratio T-F mask can be extracted *a priori* based on the *a priori* energy ratio $R(\omega,t)$ between target and interference, which is defined as follows:

$$R(\omega,t) = \left[\frac{|S(\omega,t)|^2}{|S(\omega,t)|^2 + |N(\omega,t)|^2}\right], \quad (2)$$

where $S(\omega,t)$ and $N(\omega,t)$ are the target and noise spectral values at the better ear (the ear with higher SNR). In addition, an ideal binary mask assigns the label 1 to those T-F units whose value of R exceeds 0.5 and assigns the label 0 otherwise.

As seen in Fig. 1, for mixtures of multiple sound sources there exists a strong correlation between the energy ratio and the estimated ITD and IID. The scatter plot in Fig. 1A shows the distribution of ITD and *R* for a frequency bin centered at 1 kHz. Similarly, Fig. 1B shows the results for IID at 3.4 kHz. The data is obtained from the training set in [8] consisting of 10 speech signals from the TIMIT database. For the estimation of the ideal binary mask, we employ nonparametric classification in the joint ITD-IID feature space as used in [8]. In order to estimate the ratio mask, we use ITD below 3 kHz and IID at higher frequencies. At each frequency bin, a mean curve is fitted to the distribution of the ITD/IID and *R* after removing

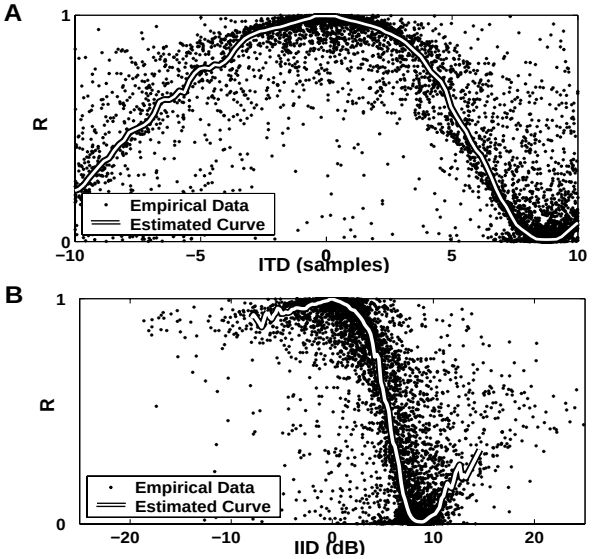


Figure 1: Relationship between ITD/IID and the energy ratio *R*. Statistics are obtained with target in the median plane and interference on the right side at 30°. (A) The scatter plot for the distribution of *R* with respect to ITD for a frequency bin at 1 kHz. The solid white curve shows the mean curve fitted to the data [see Section 2]. (B) Corresponding results for IID for a frequency bin at 3.4 kHz.

the outliers (outside of 0.2 distance from the median). Thus, for a given $I\hat{T}D|I\hat{I}D(\omega,t)$, the estimated energy ratio $\hat{R}(\omega,t)$ is the corresponding value on the mean curve.

3. Recognition strategies

We evaluate the binaural segregation system described in Section 2 as the front-end for two different recognizers. While the conventional ASR uses MFCCs as feature vectors, the missing data recognizer uses log spectral energy. A HMM toolkit, HTK [12] is used for training of both recognizers and for the testing of the conventional ASR. During testing with the missing data recognizer, the decoder is modified to incorporate the missing data methods.

3.1. The conventional speech recognizer

We use the standard HMM based speech recognizer trained on clean speech to model each word in the vocabulary (Section 4). Observation densities are modeled as mixture of Gaussians with diagonal covariance. The input to this ASR is the estimated speech spectral energy $|\hat{S}(\omega,t)|^2$.

$$|\hat{S}(\omega,t)|^2 = |X(\omega,t)|^2 \cdot \hat{R}(\omega,t), \quad (3)$$

where $|X(\omega,t)|^2$ is the spectral energy of the signal at the better ear (see Section 4). From the estimated speech spectra, we compute the MFCCs. MFCCs are chosen as feature vectors as they are most commonly used in state-of-the-art recognizers [13]. 13 Cepstral coefficients along with delta and acceleration coefficients are extracted each frame, including the $0th$ order cepstral coefficient. Frames are extracted as described in Section 2. A first-order preemphasis coefficient of 0.97 is applied to the signal.

3.2. The missing data speech recognizer

The missing data recognizer [5] makes use of spectro-temporal redundancy in speech to recognize a noisy signal based on its speech dominant T-F units. Given an observed speech vector Y, the problem of word recognition is to maximize the posterior $P(\omega_i|Y)$, where ω_i is a valid word sequence according to the grammar for the recognition task. When parts of Y are masked by noise or other distortions, Y can be partitioned into its reliable and unreliable constituents as Y_r and Y_u. In the marginalization method, the posterior probability using only the reliable constituents is computed by integrating over the unreliable ones [5]. If Y represents spectral magnitude and sound sources are additive, the unreliable parts can be constrained as $0 \leq Y_u^2 \leq Y^2$. This bounded marginalization method is shown in [5] to have a better recognition score than the simple marginalization method, and is hence used in all our experiments employing the missing data recognizer. Feature vectors for the missing data recognizer comprise of log compressed spectral energy coefficients of the signal. To provide this recognizer with the reliable and unreliable T-F units during decoding, we use the estimated binary mask as described in Section 2.

4. Evaluation results

To compare the effects of vocabulary size on the two recognition approaches outlined above, we choose two task domains. The first task is speaker-independent recognition of connected digits. This task is used in [5]. Thirteen (1-9, silence, short pause between words, zero and oh) word-level models are trained for both recognizers. All except the short pause model have 10 states, whose output distribution is modeled as a mixture of 10 Gaussians [5]. The short pause model has only three states. The TIDigits database's male speaker data is used for both training and testing. Specifically, the models are trained using 4235 utterances in the training set of this database. Testing is performed on a subset of the testing set consisting of 232 utterances from 3 speakers different from the speakers in the training set. The signals in this database are sampled at 20kHz.

The second task is the speaker-independent recognition of command and control type phrases. Two hundred and eight (206 words, silence and short pause between words) word-level models are trained for both recognizers. This task allows us to increase the vocabulary size from 13 to 208, which is a natural progression in testing the effect of vocabulary size on the recognizers. All except the short pause model have 10 states, whose output distribution is modeled as a mixture of 8 Gaussians. The short pause model has only three states. The number of Gaussians in the mixture is slightly lower compared to the first task due to lack of adequate training data for some of the models. The digital data subset of the Apple Words and Phrases database from OGI is used for both training and testing. More specifically, speakers with IDs 21 through 2604 are used for training, corresponding to 63835 utterances. 454 utterances from speakers with IDs 4 through to 19 are used for testing. The signals are sampled at 8kHz.

The two tasks also differ in perplexity. Perplexity is one indicator of difficulty of the recognition task along with vocabulary size [13]. For the digit recognition task, the perplexity is 11.0. For the command and control task the perplexity is 3.05. We calculate the perplexity empirically from the word level lattice [12]. The lower perplexity for the second task is due to the use of a restrictive grammar for this task. To test the robustness of the two recognizers in the aforementioned tasks, noise is added at a range of SNRs from -5 dB to 10 dB in steps of 5 dB. Higher positive values of SNRs are not explored, as one of the recognizers saturates to ceiling performance at 10 dB. The noise source for both recognition tasks is the factory noise from the NOISEX corpus, which is also used in [5]. The factory noise is chosen as it has energy in the formant regions, therefore posing challenging problems during recognition. It is also impulsive, making it difficult to estimate its spectrum using spectral subtraction methods [5]. In all our experiments, the target speech source was in the median plane and the noise source on the right side at 30°, making the left ear the better ear in terms of SNR.

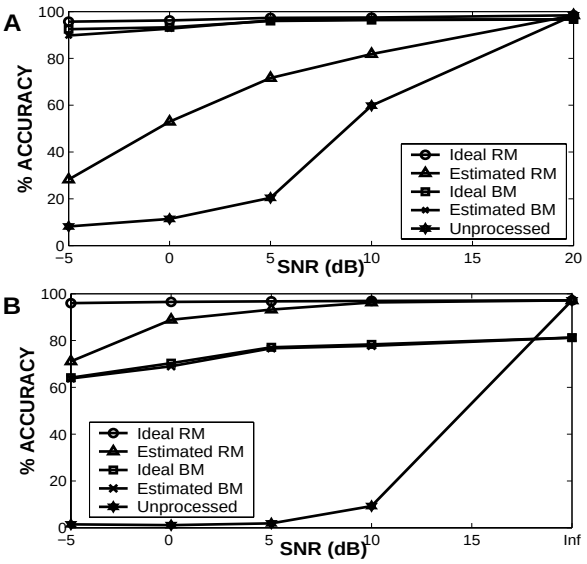


Figure 2: Summary performance of the conventional and missing data recognizers. (A) The performance of recognizers with respect to SNR on the digit recognition task. (B) Their performance with respect to SNR on the command and control phrases' task. Ideal RM refers to the performance of the conventional ASR using the ideal ratio mask. Estimated RM refers to its performance when using the estimated ratio mask by the binaural front-end. Ideal BM refers to the performance of the missing data ASR using the ideal binary mask. Estimated BM refers to the performance of the same when using the estimated binary mask by the binaural front-end. For comparison, the performance of the conventional recognizer without the use of any front-end processing is also shown.

Fig. 2A summarizes the accuracy of the two recognizers on the digit recognition task. Only a minor performance degradation is observed when the ideal ratio mask is used as a front-end for conventional ASR. The performance of the conventional ASR with the estimated ratio mask degrades faster than with the ideal ratio mask. This indicates that the conventional ASR is sensitive to the errors in our estimation of the ideal ratio mask, though the performance is substantially better than that with no preprocessing. As reported in [5], the performance of the missing data recognizer degrades very little with decreasing SNRs, indicating the adequacy of recognition using a binary mask for this task. Also, the performance with the estimated binary mask is close to that with the ideal binary mask, indicating the high quality of the front-end to estimate the ideal binary mask (see also [8]). Note that the performance of the missing data recognizer is close to the performance of the conventional ASR using

the ideal ratio mask.

Similarly, Fig. 2B summarizes the performance of the two recognizers on the task of recognition of command and control phrases. The relative performance of the two recognizers reverses with an increase in the vocabulary size. As in the digits recognition task, the performance of the conventional ASR using the ideal ratio mask is close to the ceiling performance. Additionally, its performance using the estimated ratio mask is close to the that with the ideal ratio mask, especially at SNR > 0 dB. The increased accuracy of the conventional ASR using the estimated ratio mask compared to its performance on the digits recognition task is due to the lower perplexity of this task. Its performance now is substantially better than that of the missing data recognizer using both ideal and estimated binary masks, particularly at SNR ≥ 0 dB. Lower accuracy of the missing data recognizer, under clean speech condition, on this task may be due to the use of mixtures of Gaussians with diagonal covariance. This is known to inadequately represent the observed spectral vectors [5] and this problem gets exacerbated with an increase in vocabulary size. Hence, the difference between the accuracy of conventional and missing data recognizers increases with an increase in vocabulary size (see also [7]).

5. Discussion

Our study suggests that optimal preprocessing strategies for robust speech recognition may be tied to the vocabulary size of the task. For small vocabulary applications, computation of the ideal T-F binary mask may be desirable (also reported in [11]), whereas a ratio mask may provide an improved performance with increased vocabulary sizes. When vocabulary size increases, the acoustic model space becomes densely populated. Recall that the performance curve for the missing data recognizer is steeper on the second task compared to the first task. This behavior is opposite to that of the conventional ASR, which performs better on the second task by utilizing the lower perplexity of the language model. The missing data recognizer is unable to do so. This may be caused by the inability of the missing data recognizer to represent all the speech models adequately. The log-spectral representation may have a limited expressibility in terms of distinct words that can be uniquely represented. The TIDigits database has a small vocabulary. The Applewords database with a larger vocabulary creates many more competing models during decoding. Thus, within the same T-F grid, an increased number of words need to be discriminated. With the use of a binary mask, only a smaller portion of the total T-F acoustic model space is utilized during recognition making it difficult to differentiate between competing hypothesis.

To summarize, we have presented a ratio T-F mask as a front-end for robust speech recognition using a conventional ASR. This mask is estimated using a binaural processor and provides substantial improvement over the baseline performance over a range of SNRs and on different vocabulary sized tasks. Importantly, on the larger vocabulary task, its performance with the conventional ASR is substantially better when compared to the missing data recognizer. Unlike the model-based approach in [7], the estimation of our ratio mask is independent of the signals used in the training of the speech recognizers. Hence, it is applicable even when samples of clean speech are unavailable.

Although our estimated T-F ratio mask provides promising results, other approaches for the estimation of this mask could also be explored; should a parametric curve be suspected, the parameters could be optimized to minimize recognition errors. Future work will also extend to large vocabulary tasks and explore the robustness of the binaural front-end to changes in location and number of noise sources.

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